

Curriculum Vitae

510.326.8325
ccolatos@gmail.com
<https://chriscolatos.com>
[linkedin.com/in/colatoschris](https://www.linkedin.com/in/colatoschris)
San Francisco Bay Area

Professional Summary

Innovative creative arts education leader with a proven record of growth and success in institutional leadership, student-centered active learning experience design, faculty development, and competency-based education. In roles such as Vice President for Academic Affairs, Chief Learning Officer, Director of Education, and North America Director of Learning Design, I have led large-scale institutions and programs for thousands of students, faculty, and employees, successfully developing and implementing effective on-ground and online learning strategies that significantly enhance engagement and success. Career portfolio includes creative, academic, and business acumen across multiple institutions, organizations, and colleges. I have built strategic partnerships at internal and external levels, navigating federal, state, and local regulatory environments, ensuring accreditation, and fostering community connections. My experience includes overseeing cross-functional international teams supporting thousands of artists, clients, students, and stakeholders. I am dedicated to designing and implementing data-driven, scalable professional development programs aligned with institutional goals and committed to advancing educational quality and inclusion.

Specialties: Learning Innovation & Leadership | Communication | Leadership Development | Executive Development | Professional Development | Corporate Training | Learning Experience Design | Strategic Leadership in Faculty Development & Onboarding | Competency-Based Education & Andragogy | Online Teaching Strategies & Technology Integration | Quality Assurance & Continuous Improvement in Instructional Practices | Data-Driven Decision-Making | Professional Development Program Design & Implementation | Stakeholder Collaboration & Change Management | Communication & Engagement in Diverse Academic Environments | Learning Experience Design for Online Learning | Operational Management of Faculty Processes & Systems | Career Development | AI Tools | Digital Transformation | Accessibility and Inclusion

Skills

Educational Leadership and Development:

- Strategic Growth Leadership in Education
- Customized Curriculum Design for Educational Institutions
- Competency-Based Education and Direct Assessment
- Comprehensive Onboarding Processes and Continuous Improvement
- Faculty Development, Training, Support, and Retention
- Pedagogical Innovation and Quality Assurance in Learning Programs
- Career Advising and Purpose Alignment

Learning Experience Design and Technology:

- Digital Transformation
- Learning Experience Design and Technology Integration
- Instructional Design for Multimodal Learning

- Data-Driven Learning Analytics and Improvement Plans
- Digital Accessibility
- eLearning, Hybrid Solutions, and Adaptive Learning Platforms
- Articulate Storyline 360, Captivate, Vyond, Camtasia
- Canvas, LTI, SCORM, Learning Management Systems, Instructional Technology
- Action Mapping, ADDIE, SAM, Kirkpatrick

Creative and Technical Expertise:

- AI Tools for Research, Learning, and Creative Production
- Music, Dialogue, and Sound Design for Interactive Media
- Audio Production Hardware and Software, Logic Pro X, Unreal Engine
- Google Workspace, Microsoft 365, Apple iWork
- Jira, Monday, Asana, Trello, Slack, Zoom

Project Management and Collaboration:

- Cross-Functional Project Management for Education & Corporate Programs
- Strategic Stakeholder Collaboration and Client Solutions Development
- Comprehensive Process Optimization and Continuous Improvement
- Efficient and Effective Project Actualization
- Fostering Collaboration and Building Consensus
- High Morale and Team Efficiency
- Excellent Communication and Presentation Skills
- Leading High-Performing, Agile Teams

Education

Doctor of Education (EdD), Education Policy, Organization, and Leadership

(ABD)

[University of Illinois Urbana-Champaign, College of Education](#)

Champaign, IL, USA

Human-AI Collaborative Pedagogy: Orchestrating Metacognitive Learning Ecosystems

Master of Science (MSc), Digital Education (Honours)

2017

[The University of Edinburgh, Moray House School of Education, College of Arts](#)

Edinburgh, Scotland, UK

Institutional Strategy for Online Education / Learning Engineering / Education Theory Research / Instructional Design / Learning Technologies / Learning Analytics / Educational Data Mining / Artificial Intelligence in Learning / Digital Game-Based Learning / Education Inclusion and Accessibility

Dissertation: Learning Efficacy of Multimedia in Interactive Augmented Reality Environments

Examiner: Dr. Rebecca Eynon, University of Oxford

Bachelor of Science (BSc), Sound Arts (Honors)

2006

[Ex'pression Center for New Media](#)

Emeryville, CA, USA

Electronic Composition / Interactive Audio / Multimedia Programming / Acoustics / Recording / Mixing / Mastering / Sound Design / Post Production / Signal Processing / Electronics

Learning Analytics / Educational Data Mining, Graduate Studies

2016

[Columbia University, Teachers College](#)

New York, NY, USA

Educational Data Mining / Intelligent Tutoring Systems / Artificial Intelligence in Human Learning

Intelligent Audio Systems, Post-baccalaureate non-degree Stanford University, Center for Computer Research in Music and Acoustics (CCRMA) Stanford, CA, USA Music Information Retrieval / Data Mining / Human-Computer Interaction / Machine Learning	2011
Creative Coding, Post-baccalaureate non-degree Gray Area Foundation for the Arts San Francisco, CA, USA Creative Coding / Human-Computer Interaction / Music Information Retrieval / Interactive Applications	2011
Algorithmic Computer Music (WACM), Certificate University of California, Santa Cruz Santa Cruz, CA, USA AI Music / Director, David Cope	2007
Apple Certified Master Trainer Certification, 28 Certifications Apple Inc. Boston, MA, USA - Apple Certified Master Trainer Logic Pro X 10.4, 10.3, 10.2, 10.1, 10.0 - Apple Certified Pro End User Logic Pro X 10.4, 10.3, 10.2, 10.1, 10.0 - Apple Certified Master Trainer Logic Studio 9 - Apple Certified Trainer Logic Pro 9 Adv - Apple Certified Trainer Logic Pro 9 - Apple Certified Master Pro Logic Studio 9 - Apple Certified Pro End User Logic Pro 9 Adv - Apple Certified Pro End User Logic Pro 9 - Apple Certified Master Trainer Logic Studio 8 - Apple Certified Trainer Soundtrack Pro 2 - Apple Certified Trainer Logic Pro 8 & Logic Pro 8 Adv - Apple Certified Master Pro Logic Studio 8 - Apple Certified Pro Soundtrack Pro 2 - Apple Certified Pro End User Logic Pro 8 & Logic Pro 8 Adv - Apple Certified Trainer Logic Pro 7 & Logic Pro 7 Adv - Apple Certified Pro End User Logic Pro 7 & Logic Pro 7 Adv	2006

Recent Professional Development Certifications

Leadership Communication for Maximum Impact: Storytelling — Northwestern University	2024
Computational Thinking for Problem Solving — University of Pennsylvania	2023
Data Science Foundations — Google	2023
IT Program Manager: Accessible Systems & Technology — U.S. Department of Homeland Security	2023
Section 508 Standards for Web — U.S. Department of Homeland Security	2023
Trusted Tester Web Certification Program — U.S. Department of Homeland Security	2023
Procuring Section 508 Conformant Products — U.S. Department of Homeland Security	2023
Trusted Tester Testing Tools — U.S. Department of Homeland Security	2023
Authoring Accessible Microsoft Excel Documents — U.S. Department of Homeland Security	2023
Authoring Accessible Microsoft PowerPoint Documents — U.S. Department of Homeland Security	2023
Authoring Accessible Microsoft Word Documents — U.S. Department of Homeland Security	2023
Authoring Accessible PDF Documents — U.S. Department of Homeland Security	2023
Professional Member — International Association of Accessibility Professionals (IAAP)	2023

Becoming a Product Manager: A Complete Guide — IIBA	2023
Advanced Python — LinkedIn Learning	2023
Agile Instructional Design — LinkedIn Learning	2023
ClickUp Essential Training — LinkedIn Learning	2023
Illustrator 2023 Essential Training — LinkedIn Learning	2023
Instructional Design: Needs Analysis — LinkedIn Learning	2023
Measuring Learning Effectiveness — Project Management Institute	2023

Professional Experience

Vice President for Academic Affairs

San Francisco School of Creative Arts

May 2025 – Present

San Francisco, CA

- Lead academic strategy and operations, driving initiatives that elevate excellence, improve outcomes, and advance institutional innovation.

AI Expert Contributor: Knowledge Project: Comp Sci Theory, LLMs, and Arts

Snorkel AI

February 2025 – present

Redwood City, CA

- AI Model Evaluation and Theoretical Reasoning Validation.
- Logical Error Detection and Self-Correction Mechanisms in AI.
- Data Annotation.
- LLM Prompt Engineering for Reasoning and Graduate-Level Knowledge Tasks.
- Design & curate AI reasoning tasks across CS Theory, LLMs, and Arts to strengthen model evaluation.
- Create rigorous AI benchmarking tasks to assess computational theory reasoning.

Advisory Board Member

Amotions AI

March 2024 – present

Burlingame, CA

- Advising on the direction of the AI learning and development coaching and upskilling platform.

Fellow

The Higher Education Academy

December 2016 – present

York, England

- Awarded Fellowship under the UK Professional Standards Framework (UKPSF) for excellence in teaching in higher education.

Fractional Chief Learning/Operating Officer

Unforgettable Learning

August 2018 – May 2025

San Francisco, CA

- Led corporate teams at leading global tech companies and academic teams at universities and colleges in designing and delivering data-informed, active, learner-centered teaching and learning experiences, strategies, programs, curricula, assessments, analytics, accessibility conformance, and digital transformation to optimize efficacy and engagement.
- Led the strategic design of transformative, multimodal executive learning programs to drive leadership development through customized learning journeys.
- Responsive to audience, goals, deadlines, and budget.
- Human-centered design and systematic application of evidence-based principles and methods from educational technology and the learning sciences to create engaging and effective learning experiences, support the difficulties and challenges of learners as they learn, and come to better understand learners and learning.
- Designed learning solutions, leveraging tested instructional design theories, practices, and methods to create learning activities that are engaging, activity-based, and results-oriented.
- Developed and maintained learning solutions that satisfy the needs of a diverse set of learners (instructor-led videos, online courses, quick reference guides, job aids, and performance support).
- Developed design documents, storyboards, scripts, media lists, and assessment tools.

- Translated written content into visual descriptions (graphics, animations, interactive charts and graphs, video-based scenarios) with accuracy and appropriateness.
- Drove effective project management by identifying scope, key stakeholders, and RACI.
- Delivered and facilitated pilots and train-the-trainer [T3] sessions.
- Performed learning-needs analysis and prescribed the appropriate learning experience to close performance gaps.

Chief Learning and Development Officer

July 2023 – July 2024

ASI

Santa Rosa, CA

- Designed and facilitated advanced leadership programs for team leaders, focusing on transformative leadership principles, significantly enhancing their management skills, overall team effectiveness, and morale.
- Led the onboarding and certification process, designing and facilitating human-centered orientations and integrating more than 300 new employees within the first year, enhancing their transition and engagement with company culture and processes. A testament to our welcoming culture, orientations were consistently praised as a 'life-changing experience' with an average satisfaction score of 4.94 out of 5.
- Architected and executed comprehensive organizational strategy and impactful learning programs from onboarding to leadership, including conducting more than 220 performance evaluations and resulting coaching sessions in the first year, cultivating a people-centered culture of continuous growth, retention, resilience, satisfaction, upskilling, reskilling, and knowledge, enhancing performance and capabilities through personalized feedback and development plans.
- Increased compliance by 180% within six months by auditing the certification records of every employee in the company's history and developing custom tracking software that autonomously notified employees of missing and upcoming credentials, accurately accounted for the complex and ongoing nature of compliance with individually unique certification requirements and varying timelines, all without incurring additional costs.
- Doubled historical employee retention within six months by implementing strategically targeted initiatives, maximizing limited resources and support without incurring additional costs.
- Developed SOPs and interdepartmental processes, driving digital transformation and improving organizational efficiency and efficacy through data-informed evaluation and strategic refinement.
- Empowered an inclusive, empathetic, open culture by fostering an ecosystem that nurtures growth aligned with company values.

North America Director of Learning Design

December 2020 – November 2022

SAE Institute North America

North America

- Founded the Learning Design department for the North American system of digital arts colleges from scratch.
- Created more than 400 media-rich, active learning experiences in the first six months while hiring and training a team and establishing the department from the ground up.
- Led learning designers and learning media developers in collaboration with curriculum committees, SMEs, and XFNs to develop and implement creative arts higher education teaching and active learning applications in multiple modalities for 7 North American campuses.
- Served as the institutional specialist for online and hybrid learning.
- Led academic and instructional innovation initiatives.
- Led the integration of learning design into the curriculum development of new academic programs and the enhancement of legacy programs.
- Oversaw systemwide learning management systems.
- Ensured institutional accreditation compliance and global educational quality standards.

Director of Education

October 2018 – September 2021

SAE Expression College

Emeryville, CA

- Chief academic and student affairs officer for the campus.

- Led all aspects of academic operations.
- Hired and regularly trained over 70 Program Chairs, department heads, administrators, and faculty while supporting thousands of students.
- Oversaw the Office of the Registrar, Office of Student Affairs, Library/Learning Resource Center, and all teaching and learning initiatives, facilities, and technology.
- Represented the college and built internal and external federal, state, and local regulatory, accreditation, and community partnerships.
- Created a multi-year strategy to design and implement student-centered active education programs and increase student success from matriculation to graduation and into careers.
- Developed structures for regular data reflection on organizational and student progress and identifying opportunities for action.
- Developed, implemented, and maintained internal and external policies, including Standards of Accreditation.
- SEVIS (SEVP-certified school) Principal Designated School Official (PDSO).
- Programs: Animation and Visual Effects, Game Art, Game Design and Development, Digital Filmmaking, Audio, Sound Arts, Interactive Audio, and Entertainment Business.

Academic Programs Chair

April 2010 – December 2018

Expression College

Emeryville, CA

- Directed education and operations of Audio, Sound Arts, and Interactive Audio baccalaureate programs.
- Designed and executed annual budgets and purchases of \$XXM.
- Taught courses in Audio, Game, Motion Graphics, and Film programs, including Sound Design, Post Production, Electronic Music, Game Audio, Experimental Composition, Intermedia Installation, Interactive Environment Design, Recording Arts, Signal Processing, Technical and Creative Project Development, and Capstone Research and Thesis Advisor.
- Developed and managed the rollout of baccalaureate programs across campuses.
- College representative for Ed, Tech, and community panels, conferences, and workshops.
- Awarded Instructor of the Year. Awarded Faculty Member of the Year.
- Improved KPIs, including measurable student success, new enrollments, attrition, and reentries.
- Original Accreditation and renewal experience.
- Experienced in U.S. DoE Gainful Employment regulations for Title IV eligible programs.
- Experienced in Title IX regulations.
- Experienced in ADA protections and accommodations.
- Extensive policy development and execution experience.
- Managed 124,000 square feet of zero-downtime audio/video facilities and equipment across two campuses.
- Advised, planned, and supported Facilities in the design and development of the branch campus and the multi-million-dollar build-out of the main campus.

Director of Professional Training

September 2008 – June 2014

Expression College (Ex'pression PRO)

Emeryville, CA

- Independently led and developed the college's professional training and certification division from the ground up.
- Two hundred twenty-one unique students in the first year.
- In the black P&L for the complete duration of tenure.
- Authorized Training Center/Partner initiation experience. Apple Gold Level distinction.
- Courses/certifications in Apple (AATC), Avid (ALP), Ableton (ACTC), Adobe, social media, animation/visual effects, game design, filmmaking, creative coding, electronic music, synthesis, electronics, projection mapping, and installation.
- Ongoing training of 40 Pixar artists.
- Accreditation initiation and compliance.

- Hired, developed, and managed faculty and staff, as well as developed curricula, instructional design, scheduling, budget, student advising, social media, and online course development.

Director of Program Development

January 2009 – January 2012

Expression College

Emeryville, CA

- Developed new academic degree programs, including the college's most populous programs.

Professor / Academic Course Director

June 2002 – April 2010

Expression College

Emeryville, CA

- Developed and managed academic courses.
- Taught in the Audio, Game, Motion Graphics, and Film programs.

Founder and Chief Creative

July 1993 – present

THE STUDI/O

Oakland, CA

- 200+ producing, composing, recording, mixing, mastering, sound design, AR/VR, and game credits.
- AI music since 2007.
- Lead the organizational vision, strategy, and creative direction.
- Serve as the principal producer and primary liaison to support the goals of hundreds of artists and organizations.
- Drive the planning, project management, and delivery of solutions spanning music and sound, games, interactive media, AR/VR, mixed media composition, design, performance, and installation.
- Partner and produce for high-profile brands, including Apple, Google, LeapFrog, Honor Code, Legends The Game, Robot Invader, Eagre Games, OtoCast, Ableton, Universal Audio, Tarpan Studios, Twisted Tools, iZotope, PETA, Stardust Brands, Harbor Picture Company, GQ, Salvatore Ferragamo, and Vogue.
- Deliver top-quality music and audio production for globally recognized artists, including Greg Pillinganes, Siedah Garrett, Narada Michael Walden, James Taylor, Carole King, Smokey Robinson, Billy Joel, Sting, Elton John, and Katy Perry.

- Music Production
- Music Composition
- Songwriting
- Arranging
- Sound Design
- Post Production
- Field Recording
- Foley

- Dialogue
- ADR
- Mixing
- Mastering
- Games
- Interactive Media
- AR/VR/XR
- Mobile

Clients include:

- | | |
|--------------------|------------------------------|
| ▪ Apple | ▪ City of Emeryville |
| ▪ Google | ▪ WorkforceLogic |
| ▪ LeapFrog | ▪ SEAMUS |
| ▪ Honor Code | ▪ Peachpit Press |
| ▪ Legends The Game | ▪ Focal Press |
| ▪ Robot Invader | ▪ KSFR Santa Fe Public Radio |
| ▪ Eagre Games | ▪ Shure |
| ▪ OtoCast | ▪ SMACNA |
| ▪ Universal Audio | ▪ 1Light Media |
| ▪ Ableton | ▪ Country Road Films |
| ▪ Twisted Tools | ▪ Harbor Picture Company |
| ▪ iZotope | ▪ Stardust Brands |
| ▪ Tarpan Studios | ▪ Alsiona |
| ▪ Bag End | ▪ Mazrine |

- Massimiliano Giornetti
- Salvatore Ferragamo
- GQ
- Playboy
- Vogue
- PETA
- PSY
- Scientist (dub pioneer)
- Greg Phillinganes
- Siedah Garrett
- Narada Michael Walden
- James Taylor
- Carole King
- Smokey Robinson
- Billy Joel
- Sting
- Elton John
- Katy Perry

Intermedia Composer, Performer, and Sound Artist

July 1993 – present

c^2 (self)

International

200+ Producing, Recording, Mixing, Mastering, Sound Design, AR/VR, and Game Credits

Selected Works & Exhibitions

[2026] CANTUS

Browser-based generative composition engine that writes voiced, phrased, and cadenced music across jazz, classical, electronic, and song idioms, builds the rhythm under it, then lets you reshape it. Five linked views: a piano-roll Composer Map with chord cards and a live harmonic-tension trace, a Harmonic Orbit of chord substitutions ranked by shared tones, a Voice Loom for voice-leading edits, a Field that lays the generative layer across the whole passage, and a Drum Room whose break engine writes the groove across more than fifty styles, steered by chaos, ornament, and polymeter. Temperament sliders for risk, color, motion, and spread, six one-tap harmonic transformations, and motif-based melody development. Two traditions of algorithmic composition run side by side. Learn is symbolic AI in the lineage of David Cope's Experiments in Musical Intelligence: it ingests your own MIDI, infers key by Krumhansl-Schmuckler profile correlation, separates voices by onset clustering, distills recurring motifs, and builds a first-order Markov model of chord-root motion, then composes fresh music in that style by rule rather than imitation. Process is its formal counterpart, six layered generative engines: Euclidean rhythm, cellular automata, L-systems, Markov chains, Xenakis-style stochastic distributions, and parametric curves, each with its own live visualizer, every note locked to the sounding chord. Multi-track Standard MIDI File export, OfflineAudioContext WAV rendering, and sample-accurate scheduling on the Web Audio clock. Route live Web MIDI, each part on its own channel, to your synths or a DAW to hear it at full resolution.

[2026] Dynamic Stochastic Synthesis

In Suspended Harmony

Composition for 20 DSS voices revealing the overtone series of a single fundamental. Partial 1 through 16 emerge across nine harmonic regions, building from a bare octave to full spectrum before dissolving back to the fundamental, altered by a ghost of the 15th partial.

Lemniscate

Realization of DSS with no key, no scale, and no harmonic plan. Twenty voices are defined only by frequency barriers in raw Hz. The random walk determines every pitch. Consonance is a statistical event.

Gendy c2

Sparse realization faithful to Xenakis's GENDY3. One to five voices play at any moment, exposing each waveform mutation as a naked, audible event.

Chiaroscuro

Composition modulating through six tonal centers across twelve regions. Four voice classes rotate leadership, eleven asymmetric breath-swells shape the dynamics, and Cauchy bursts punctuate harmonic shifts as structural events.

Ergodic

Composition built on durational structure rather than harmonic structure. No tonal center. Twenty voices span 18 Hz to 14 kHz, untethered from harmonic function; frequency spread itself becomes compositional material, narrowing to near-pure tones in silence and exploding beyond defined ranges during twin eruptions. Twelve sections follow Fibonacci-adjacent proportions across 577 seconds. Full Cauchy distribution, maximum walk rates, density from a single sub-bass voice to all twenty at once. The form is symmetrical: void to void, with two violent climaxes separated by a shattering collapse.

Compositions share a single engine: a recreation of Xenakis's Dynamic Stochastic Synthesis in a Web Audio API AudioWorklet. Breakpoints evolve at sample rate via Gaussian and Cauchy random walk with reflecting barriers. Stochastic parameters are themselves subject to second-order random walk. Signal chain: DSS voices, tanh saturation, 12-second convolution reverb, shimmer delay. No samples. No oscillators. Every sound is generated from arithmetic on breakpoint arrays.

[2026] KAOSBRK

Browser-based generative drum instrument that fuses sample-based breakbeat manipulation with algorithmic pattern synthesis. STFT-based onset detection and adaptive classification split any uploaded break into kicks, snares, ghost notes, and hats. A chaos engine performs progressive structural transformations through per-voice step swaps, half-bar mirroring, bar rotation, cross-voice substitution, and stutter injection, exposing the instrument's compositional logic as a continuously tunable parameter space. Twelve time signatures including odd and compound meters. Eleven-voice mixer with solo/mute, pan, faders, and real-time VU metering. Sidechain ducking, comb filter, bitcrush, 4x oversampled tanh drive. Click-and-drag velocity editing per cell. Standard MIDI File export, OfflineAudioContext WAV rendering, live Web MIDI output. Sample-accurate scheduling locked to the Web Audio clock.

[2026] NEUROTRKR

Nine-channel tracker-style step sequencer and synthesizer running entirely in the browser. No samples. Every kick, snare, bass, and pad is synthesized live from oscillators, noise, and waveshaping via the Web Audio API. Five drum and four melodic channels, each with multiple synthesis variants, variable time signatures including odd meters, grid resolutions to 1/256, swing, delay, and convolution reverb. Per-note probability and Elektron-style trig conditions. Pattern variants (A/B/C/D) assignable per bar. A preset library of 135 rhythms across 18 genres, from the jungle breaks to breakcore, techno, IDM, and artist references (Aphex Twin, Venetian Snares, Autechre, Squarepusher). Copy/paste bars, per-channel pattern shift, one-level undo, and autosave to browser storage. Sample-accurate scheduling

locked to the Web Audio clock, Web Worker-driven background playback, Standard MIDI File export, and live Web MIDI output to external hardware and software. Mobile and desktop.

Two UI skins, including a retro OctaMED tribute.

[2025] d a s h b o a r d s

[2024] c4lcul470r mu51c

[2023] c^2 , Discordant Arcana

[2021] Society for Electro-Acoustic Music in the United States

Adjudicator for SEAMUS 2020 and 2021 National Conferences

[2019] OtoCast Mobile App, The City of Emeryville

Content Producer/Sound Designer and Audio Engineer

[2019] ZED, Eagre Games

Multiplatform/VR, Recording Engineer, Sound Designer, VO Editor, Dialog Processing /
Mastering

[2018] Dead Secret Circle, Robot Invader

Multiplatform/VR, Recording Engineer, Sound Designer, VO Editor, Dialog Processing /
Mastering

[2017] Narcosis, Honor Code

Multiplatform/VR, Audio support

[2017] c^2 , Stratum, Center for New Music, San Francisco, CA

[2016] Salvatore Ferragamo, Composer, Lo Splendore Della Vita

[2016] St0cHastAk0v1Ch, autocomposer

A fully programmable sequencer application developed in a graphical environment.

Probability distribution autocomposer functionality with Dmitri Shostakovich homage-scale constraint. Controls for independent and global pitch and velocity, independent and vector linear pitch and modulation, independent and global note on/off, and quick-drum-sequencer section

[2015] c^2 , Electroacoustic Compositions for Guitar featuring Mike Keneally

[2015] c^2 , Nightingale, sonification of Phaedrus

Plato's dialog Phaedrus, represented by sound and musical traits; morphology; phonology.

[2015] Skeuomorph, algorithmic concatenation

Atmospheric story generated from a narration library employing corpus-based concatenative synthesis

[2014] Designing Musical Software, San Francisco State University, A.S. Theater

Presentation of live audio coding in a graphical environment, followed by an interactive performance utilizing original music software

[2014] Earthquakes (Remix), The Candle Thieves

[2014] Simulacrum, formant transformations with Kyma

[2013] Face2Phase, facial gestural controller for audio and visual synthesis

[2013] PSY, Composer, Vogue Italia feature

[2013] c^2 , Boulevard, Chicago, IL

Electronic performance, laptop, physical objects, electronics, controllers

[2012] Adaptive Systems Toolkit

Development in Kismet, graphical environments for interactive design

[2012] Katy Perry, algorithmically produced composition, Vogue Italia

[2011] GQ, Composer

- [2011] Laptop Quintet, Impromptu, CCRMA, Stanford University, Stanford, CA
- [2011] PETA, Composer, Save the Seals campaign
- [2010] Playboy, Composer, features
- [2010] SynthesImage, spectral synthesis, audio sonification of imagery
- [2010] Twisted Tools: Sound Designer and Drum Programmer
 - Vortex, Scapes, Rolodecks, S-Layer, and Buffeater instruments
- [2009] CodeTrain, algorithmic analysis and performance
 - Algorithmic compositional tool coded in LISP to create new performances in the style of John Coltrane. The software is designed to recognize patterns in rhythm, note order, and melodic, harmonic, and contrapuntal movements and progressions, producing new music in the style of the recorded data set
- [2008] Glitch Machine v1.2
 - LISP-coded system used to analyze and modulate rhythm data sets and create rhythmically new, complex, and polyrhythmic patterns
- [2008] c^2 , Tonic Room, Chicago, IL
 - Electronic performance, laptop, physical objects, electronics, controllers
- [2007] a-morph-ous, algorithmic composer
 - Transformative algorithmic compositional software written under the mentorship of David Cope and his Experiments in Musical Intelligence, with the original intent to produce new symphonic works in the style of Gustav Mahler. Developed as a model, the software recognizes patterns in rhythm, note order, and melodic, harmonic, and contrapuntal progressions, producing new music in the style of the input data set. Developed as three analysis systems: polyphonic without rhythm for intractability, polyphonic with rhythm utilizing a Markov decision model, and polyphonic with rhythm and measure structure using a Markov model while constraining analysis to smaller segments for less divergent composition
- [2007] eLiS, list sonification
 - Data analysis tool written in Common LISP. Intentioned initially to provide an auditory representation of inaudible information such as earthquake activity, populations across locations, weather patterns, and extraterrestrial storms. eLiS scales and translates lists of information into MIDI pitch, velocity, on-times, duration, and channels, among other parameters, to observe audibly
- [2006] IMPULSION — electroacoustic improvisation & field recording soundscape
 - Multi-channel installation of an improvised performance of a self-designed percussion instrument and found sounds processed by discrete impulse responses captured at their original recording location - autonomously generated, the signals are processed to effect perception
- [2006] Legends The Game, complete sound design
- [2006] Viriconium, Muse Gallery, Chicago, IL
 - Autonomous and interactive prepared percussion and synthesis installation utilizing a recursive feedback instrument
- [2005] Girl on a Train, McGonigal's, Barrington, IL
 - Live electronics, sound, and light experiments with live electronic ensemble 7 League Boots
- [2005] Scientist, Dub Producer Hopeton Brown, Engineer

- [2004] Colour and Silence. The Onion, Chicago, IL
Electronic performance, live electronics, sound and light experiments, noisemakers, controllers, drum kit
- [2004] Explorer Smart Globe, LeapFrog, Sound Designer and Editor
- [2004] String dissection
Interactive music concatenation system developed within a game engine.
- [2003] c^2 , Lilly's, Chicago, IL
Electronic performance, laptop, physical objects, electronics, controllers, drum kit
- [2002] Rock for the Rainforest, Rainforest Foundation Concert, Carnegie Hall, Arranger: Billy Joel
- [2002] Rock for the Rainforest, Rainforest Foundation Concert, Carnegie Hall, Arranger: Carole King
- [2002] Rock for the Rainforest, Rainforest Foundation Concert, Carnegie Hall, Arranger: Elton John
- [2002] Rock for the Rainforest, Rainforest Foundation Concert, Carnegie Hall, Arranger: James Taylor
- [2002] Rock for the Rainforest, Rainforest Foundation Concert, Carnegie Hall, Arranger: Smokey Robinson
- [2002] Rock for the Rainforest, Rainforest Foundation Concert, Carnegie Hall, Arranger: Sting
Reconceptualization of previously composed work. Worked with Musical Director Narada Michael Walden on high-profile conversancy benefit spearheaded by Sting and Trudie Styler
- [2002] Siedah Garrett (Michael Jackson/Madonna), Engineer, Siedah
- [2001] c^2 , About Time, tape effects study
- [2000] Jacked Kerowack
Composed by transforming recordings of Jack Kerouac reading his work. Conversely, the spoken word is affected by parameters unique to the newly generated composition
- [1999] इंडिया, field recording library created throughout Australia and India
- [1998] Duets, J. Angelo (saxes, flute) C. Colatos (drums/electronics) live 90.7 FM SFPR
- [1997] Blight, Cerrillos Historical Society, Duet, w/ J. Angelo. Cerrillos, NM
- [1997] Inharmonique, Michael Wilding Gallery, Santa Fe, NM
Installation, drone, two reel-to-reel tape machines and electronics, experiments in equal temperament, alternative tunings, and beating
- [1997] Improvised processed drone ensemble. Whispering Dove, Santa Fe, NM
- [1996] Percussion Ensemble, International Community School, Addis Ababa, Ethiopia
- [1996] አጫጫታ, field recording library created throughout East Africa
- [1996] Baroque Ornaments, Santa Fe Art Institute, Santa Fe, NM
Artist-in-residence. Soundscape installation for tape and projection of reimagined and recorded historical and contemporary local sound.

Master Trainer (ACT)

Apple, Inc.

October 2006 – September 2021

Boston, MA / North America

- Apple Certified Master Trainer.
- Trained and Certified thousands of artists, professionals, and technologists in Apple software and hardware.

North America Chair of Curriculum Design

SAE Institute North America

November 2015 – March 2019

North America

- Led North American team of instructional designers and SMEs in original program development.
- Led institutional strategy, compliance, market relevance, articulation, curricula, resources, active teaching and learning tools, and online platform initiatives.
- Developed Diplomas, Associate's, and Bachelor's degrees for nine colleges.

Arts Program Developer (Volunteer)

After-School All-Stars

January 2018 – January 2019

Oakland, CA

- Developed comprehensive after-school arts programs in underserved communities.

Guest Lecturer

2018

University of Colorado Denver, MEIS Department in the College of Arts & Media

Denver, Colorado

- Taught an intensive game development and algorithmic music course using the Unreal Engine.

Lecturer and Learning Designer

September 2012 – November 2015

San Francisco State University

San Francisco, CA

- Lecturer and Learning Designer for Graduate and Undergraduate level courses.
- Taught: Sound Design, Advanced Recording & Production, Post Production, Electronic Music, and Game Audio.
- Student review average: 9.73/10.

Producer, Composer, and Music Supervisor

February 2011 – August 2015

Country Road Films

Los Angeles, CA

- Multimedia production, mastering, composition, and performance for clientele including GQ, Playboy, Vogue, PSY, and Katy Perry.

Executive Director

February 2011 – April 2015

Apple Training Hawaii

Kilauea, HI

- Founded professional certification training centers from the ground up.
- In-the-black P&L from inception and growth for the complete duration of tenure.
- Oversaw academics and operations.
- Designed programs, courses, curricula, instructional tools, and policies.

SEO QR

November 2008 – October 2010

Google

Mountain View, CA

- Search Quality Optimization
- User Representation/User Intent
- Task Language & Location
- Multiple Meaning Queries
- Action, Information, and Navigation — “Do-Know-Go”
- Internet Safety & Prevention of Doorway Pages/Sneaky Redirects

Producer, Songwriter, Arranger, Recording, Mixing, and Mastering Engineer

July 2001 – September 2004

Narada Michael Walden - Tarpan Studios

San Rafael / Los Angeles, CA

- Worked closely with one of Billboard Magazine’s Top Ten Producers of All Time: 57 #1 Hits, multi-GRAMMY, Emmy & multi-Platinum, Producer and Artist Narada Michael Walden.
- Worked on countless projects, including work for artists Greg Phillinganes, Siedah Garrett, James Taylor, Carole King, Smokey Robinson, Billy Joel, Sting, and Elton John.

“A man for all seasons...from Ahmad Jamal to the hippest in contemporary music, Chris knows how it's done...a whiz with the computer and the drums. Chris has the genius touch.” – Narada Michael Walden

On-Air Host and Show Producer

September 1997 – August 1998

KSFR Santa Fe Public Radio

Santa Fe, NM

Selected Broadcasts

- [05.31.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Steve Reich
- [05.24.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Grisey and Murail
- [05.17.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Alexander Scriabin
- [05.10.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Morton Feldman

[05.03.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Daphne Oram
 [04.26.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Prince Far I
 [04.19.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: King Tubby
 [04.12.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: François Bayle
 [04.05.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Pierre Schaeffer
 [03.29.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Edgard Varèse
 [03.22.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Éliane Radigue
 [03.15.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Laurie Spiegel
 [03.08.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Pierre Boulez
 [03.01.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Bernard Parmegiani
 [02.22.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Mathews and Risset
 [02.22.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Pauline Oliveros
 [02.15.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Earle Brown
 [02.08.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: David Bowie
 [02.01.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: R. Stevie Moore
 [01.25.98] Co-Host KSFR 90.7 FM, 'Stereo Jelly,' Punk, Noise, Avant-Garde
 [01.25.98] Host KSFR 90.7 FM, Santa Fe Public Radio, 'All Talk Don't Bounce'
 [01.25.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: La Barbara and Monk
 [01.18.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Coltrane, Jones, Ali
 [01.11.98] Co-Host KSFR 90.7 FM, 'Stereo Jelly,' Punk, Noise, Avant-Garde
 [01.11.98] Host KSFR 90.7 FM, SF Public Radio, Eclecticism: Reggae, Rap, Jazz, Electronic, World
 [01.11.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Pérotin and Arvo Pärt
 [01.04.98] Co-Host KSFR 90.7 FM, 'Stereo Jelly,' Punk, Noise, Avant-Garde
 [01.04.98] Host KSFR 90.7 FM, SF Public Radio, Eclecticism: Reggae, Rap, Jazz, Electronic, World
 [01.04.98] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Ayler, Coleman, Dolphy, Braxton
 [12.28.97] Co-Host KSFR 90.7 FM, 'Stereo Jelly,' Punk, Noise, Avant-Garde
 [12.28.97] Host KSFR 90.7 FM, SF Public Radio, Eclecticism: Reggae, Rap, Jazz, Electronic, World
 [12.28.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: The Velvet Underground
 [12.21.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Iannis Xenakis
 [12.14.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Terry Riley
 [12.07.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Scelsi and Schnittke
 [11.30.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Berg and Webern
 [11.23.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Aphex Twin
 [11.16.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Squarepusher
 [11.09.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Autechre
 [11.02.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Negativland
 [10.26.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Alvin Lucier
 [10.19.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Christian Wolff
 [10.12.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: David Tudor
 [10.05.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: La Monte Young
 [09.28.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: John Cage
 [09.21.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Messiaen and Satie
 [09.14.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: Fela Kuti
 [09.07.97] Host KSFR 90.7 FM, Santa Fe Public Radio, Artist Portrait: KRS-ONE

Professional Associations

- International Association of Accessibility Professionals (IAAP)
- Acoustical Society of America (ASA)
- Advance HE (Fellow)
- Association for Computing Machinery (ACM)
- Audio Engineering Society (AES)

- Center for New Music (C4NM)
- Fractured Atlas
- Game Audio Network Guild (GANG)
- Institute of Electrical and Electronics Engineers (IEEE)
- Interactive Audio Special Interest Group (IASIG)
- International Society for Music Information Retrieval (ISMIR)
- Society for Electro-Acoustic Music in the United States (SEAMUS)
- Society of Professional Audio Recording Services (SPARS)
- Special Interest Group on Computer Graphics (SIGGRAPH)

Academic Courses Taught

- Principles of Sound
- Studio Production
- Signal Processing
- Electronic Music Production
- Studio Maintenance and Audiovisual Technology
- Audio Post Production
- Game Audio
- Advanced Studio Production
- Sound Design
- Advanced Audio Post Production
- Audio Programming for Interactive Media
- Mastering and Media Preparation
- Audio Scripting
- Advanced Game Audio
- Interactive Design
- Interactive Audio Fundamentals
- Adaptive Audio Integration
- Intermedia Composition
- Basic Recording
- History of Music Production
- Computer Music Production
- Intermediate Recording and Production
- Digital Audio Workstation
- Digital Audio Workstation II
- Computer Music Production II
- Intermediate Recording and Production II
- Audio Post Production
- Advanced Recording and Production
- Advanced Recording and Production II
- Audio Post Production II
- Studio Etiquette and Psychology
- Capstone Project (Thesis)
- Add'l conferences, panels, and lectures

Recent Volunteer Experience

Society for Electro-Acoustic Music in the United States (SEAMUS), Adjudicator 2019 – 2021
Judge for SEAMUS 2020 & 2021 National Conferences.

Pride Arts and Crafts Social, Facilitator 2019 – 2021
Providing a safe and inclusive space for creative expression and community building for LGBTQIA+ students.

AfroComicCon Festival, Organizer 2020 – 2021
Platform to showcase and empower artists historically denied equal access to opportunity and promote diversity in the creative arts.

After-School All-Stars, Arts Program Developer 2018 – 2019
Developed comprehensive after-school arts programs in underserved communities.

Audio Engineering Society, Faculty Liaison, AES Student Chapter 2016 – 2019

Girls Who Code, Facilitator 2017 – 2018
A mission to close the gender gap in technology; to change the image of what a programmer looks like & does.

Girls Make Beats, Organizer 2016 – 2017
A mission to close the gender gap in music education and technology.

Selected Writings

WHOSE CHILDHOOD DOES ELIMINATION PROTECT? A CRITICAL REVIEW OF THE ILO FRAMEWORK FOR ACTION ON CHILD LABOUR 2023–2025 2026

This review critically examines the International Labour Organization's Framework for Action on Child Labour 2023–2025, the ILO's operational response to the first recorded rise in global child labour in two decades. It argues that the framework globalizes not a neutral protective norm but a particular model of childhood: a chronological age line joined to compulsory schooling as the sole legitimate site of learning, diffused through statistical categories, ratification campaigns, corporate due diligence, and trade clauses, and constructed so that the children it governs have no standing to contest it. The framework's measurement standard classifies as pathology the family-based work that anthropology documents as pedagogy; it omits the light-work category of its own conventions; it makes elimination mandatory while leaving social protection aspirational; and its survivor-informed evidence base can register harm but not benefit, despite economic evidence that poorly sequenced prohibition deepens household poverty. The review proposes participation-informed policymaking with standing for organized working children, social protection sequenced as a precondition of withdrawal, and hazard rather than work as the object of elimination: criteria against which the framework's 2026 Marrakech successor can now be judged.

STUDENTS AS DESIGNERS OF MEANING: MULTILITERACIES, IDENTITY, AND THE TRANSFORMATION OF ENGAGEMENT IN ARTS HIGHER EDUCATION 2026

Arts higher education faces a persistent engagement paradox: students in future-oriented creative disciplines often disengage not from content, but from pedagogy. While curricula foreground innovation, instruction frequently remains transmission-based, positioning learners as recipients rather than designers of knowledge. Drawing on a critical synthesis of contemporary scholarship, this paper argues that repositioning students as designers of meaning constitutes a pedagogical reframing. Grounded in the multiliteracies framework, the analysis conceptualizes engagement as a function of agency, identity formation, and multimodal design. When learners construct meaning across media, situate their work within lived identities, and participate in authentic creative workflows, engagement shifts from a motivational variable to a consequence of instructional design. Examining multimodal project-based practice, identity-affirming pedagogy, and AI-integrated creative production, the paper traces both the transformative potential and the structural constraints of this shift. Faculty resistance, uneven digital self-efficacy, institutional inertia, and limited professional learning infrastructures complicate redesign, underscoring that pedagogical transformation requires not only new tools but new epistemological commitments. The central contribution is a theoretical synthesis demonstrating that multiliteracies, aligned with AI-mediated creative workflows, offer a coherent framework for reimagining arts education. Integrating active learning, multimodal meaning-making, and identity development within contemporary creative technologies prepares students not merely to use generative systems, but to shape them with critical, creative, and ethical intent.

GENERAL-PURPOSE COGNITIVE INFRASTRUCTURE IN CREATIVE ARTS HIGHER EDUCATION 2025

This paper introduces General Purpose Cognitive Infrastructure (GPCI) as a conceptual framework for understanding how AI systems function as foundational cognitive resources in higher education in the creative arts. Synthesizing cognitive load theory, distributed cognition, and research on human-AI co-intelligence, the paper positions AI not as a substitute for creative thinking but as infrastructural support that redistributes cognitive demands across learners, tools, and communities. The framework specifies how AI can reduce extraneous load, modulate intrinsic complexity through adaptive scaffolds, and protect germane load for schema formation, aesthetic judgment, and iterative creative problem solving. Through analysis of machine learning education for creative practitioners and AI-assisted creative workflows, the paper illustrates how GPCI principles appear in practice, including example-based interactive learning, dialogic intent mediation, and rapid feedback cycles. The analysis argues that effective curriculum integration requires staged pathways that build AI literacy alongside domain craft, explicit attention to the redistribution of cognitive load, and assessment approaches that foreground process evidence and learner agency rather than product alone. The paper concludes by outlining implications for curriculum design, faculty

development, and institutional governance, emphasizing equity, transparency, and the preservation of human creative agency within AI-augmented studio ecosystems.

SYMBIOTIC INTELLIGENCE IN CREATIVE ARTS EDUCATION: NAVIGATING THE PARADIGM SHIFT TOWARD HUMAN-AI CREATIVE PARTNERSHIP

2025

This literature review synthesizes scholarship on human-AI collaboration in creative arts education across sound art, music production, filmmaking, and digital media, with attention to how generative tools reshape creative learning, authorship, and assessment. The analysis identifies a central tension: AI can accelerate technical proficiency and expand iterative exploration, yet may also introduce dependencies, narrow stylistic variation, and obscure evidence of human learning when outcomes are co-produced with systems. Across the literature, explicit pedagogical frameworks for AI-augmented studio learning remain underdeveloped, and existing assessment practices are often misaligned with distributed, tool-mediated creative work. In response, the paper advances a Symbiotic Creative Intelligence Framework that integrates distributed cognition and multiliteracies to reframe creative accomplishment as an interaction among learners, tools, and communities, and to emphasize process transparency, staged competency development, and cultural responsiveness. The review argues for assessment approaches that privilege traces of practice, including documentation of orchestration, iterative decision-making, critical evaluation, and aesthetic agency across AI-assisted and AI-limited conditions. The paper concludes with a research agenda calling for longitudinal studies of creative skill transfer, cross-cultural investigations of aesthetic impact and bias, and experimental validation of process-centered assessment designs. Ultimately, it positions human creative development, voice, and agency as the non-negotiable aims guiding responsible AI integration in creative arts education.

THE LIMINAL CANVAS: ENTANGLED CONSCIOUSNESS IN HUMAN-AI CREATIVE SYMBIOSIS

2025

This article interrogates the emergent ontological space where human and artificial intelligence converge in artistic co-creation, proposing a framework we term "entangled consciousness aesthetics." Drawing on quantum entanglement as metaphorical architecture, we examine how the dissolution of traditional creator/tool binaries generates novel phenomenological experiences that transcend human and computational limitations. Through rigorous analysis of collaborative artistic outputs across diverse media, we demonstrate that the most profound creative breakthroughs occur not when AI functions as a servant or master but as an ontologically distinct yet intimately connected consciousness whose capabilities complement rather than simulate human creativity. This symbiotic relationship manifests "impossible artifacts"—works that could never emerge from either intelligence operating in isolation. We argue that educational institutions must fundamentally reconceptualize creativity as an intersubjective dialogue between different forms of consciousness rather than an exclusively human capacity to be augmented by technological tools. The implications extend beyond aesthetics into ethics, epistemology, and the boundaries of what constitutes artistic identity in a post-anthropocentric creative landscape.

AUGMENTING THE REALITY OF ACCESSIBLE LEARNING ENVIRONMENTS

2023

Education Technology Insights

The integration of Augmented Reality (AR) technology holds immense potential to revolutionize accessible digital learning environments for individuals with disabilities. The framework provides opportunities for immersive, interactive, and inclusive learning experiences that enhance student engagement. The present research explores the theoretical foundations and technological advancements of AR in addressing the unique needs of people with disabilities, focusing on practical applications and prospective advances, exploring how AR can integrate visual and auditory cues for learners with hearing and sight impairments, incorporate haptic feedback and gesture recognition to facilitate tactile and kinesthetic interactions for students with motor impairments and tailor content delivery methodologies for those with cognitive disabilities. The study emphasizes the significance and efficacy of AR in catalyzing equitable and empowering online learning environments and highlights implications for all learners, enabling individuals with disabilities to participate and thrive fully.

LEARNING EFFICACY OF MULTIMEDIA IN INTERACTIVE AUGMENTED REALITY ENVIRONMENTS 2017

Augmented Reality is an incipient technology rooted in traditional multimedia with attractive applications for education. Mobile devices travel with billions of people globally, and the inherently mobile nature of AR, with its simple and inexpensive implementation relative to the potentialities exhibited for pervasive, personal, and connected learning, indicates that it is poised to supplement the future of multimodal educational dissemination. The present study explored the learning efficacy of multimedia forms within interactive augmented reality environments. The results are essential for confirming academic quality when employing emergent practices. This information may be of value for educators, instructional designers, and learners seeking to utilize empirical evidence as the basis of educational design decisions and learning strategies. The study isolated media types within AR experiences and evaluated their efficacy for learning.

EFFECTS OF ARTS EDUCATION AND UNDERSERVED COMMUNITIES 2016

This article explores the potential of arts education access and practice and advocates that arts education is integral to human rights, social justice, and intellectual, democratic, and cultural inclusivity with qualities of importance to the empowerment and equity of underrepresented communities.

ADAPTIVE LEARNING DESIGN IMPLICATIONS: EFFECTS OF MULTIMEDIA ON LEARNING EFFICACY AND COGNITIVE LOAD 2016

An experimental and data-driven study to inform effective adaptive learning domain model development decisions with an incipient exploration of cognitive load and learning efficacy when utilizing multimedia stimuli. These platforms' systems rely on sophisticated feedback from human interaction, educational theory, learning analytics, and educational data mining. The pilot comprised an overview of the proposed research, relevant literature on orthogonal and intersecting agendas, imbricated methodological experimentation, data processing, and anticipated challenges. Results demonstrate that media types and combinations of media exhibit varying cognitive load, consequently impacting learning efficacy.

APPLE PRO TRAINING SERIES: Logic Pro 9 and Logic Express 9: Professional Audio Production 2009

Peachpit Press: Technical Reviewer for Apple Pro Training Series: Logic Pro 9 and Logic Express 9: Professional Audio Production